Winch wire log and RVSS Appendix A checks: RR2403 hydro

Origin and Evolution of Intraplate Magmatism at the Revillagigedo Archipelago

vessel Roger Revelle · PI Katherine Kelley · 2024-03-29 to 2024-04-27 · winch hydro (Markey DESH-5)

Source: data/RR2403_cast · 1 files (*hydro*.txt) · format openrvdas_lci90i (LCI-90i strings logged by OpenRVDAS (ISO timestamp first))

Written by winch-data-processing 0.1.0

No check failed; 6 can't be shown met from the config (warn), 7 passed

Checks

Check	Section	Status	Detail
Deployments	A.5.2.1	info	1 found
Tension within the safe working tension	Table A.8.2, A.2.19	pass	every deployment within the SWT 4000 lbf (FS 2.5)
Never below the lowest FS allowed	Tables A.8.3, A.8.4	pass	never below FS 2 for a cable
Peak tension within the elastic limit	A.5.6	pass	never beyond FS 1.8
Slow-haul checks below FS 2.0	Table A.8.4	n/a	Table A.8.4 only (wire rope from FS 2 to 1.5); the selected FS 2.5 is in Table A.8.2
Tension logging rate	Table A.8.2	pass	every deployment logged at 3 Hz or more
Gaps in the tension record	Tables A.8.2 to A.8.5	pass	no gap over 5 s in a deployment
Tension alarm setting	Table A.8.2	warn	no [monitoring] alarm_tension; must be at or below 3571 lbf (FS 2.8)
Tension below the alarm point	Table A.8.2	warn	above Table A.8.2's alarm point, 3571 lbf (FS 2.8): cast 1: 0.1 s above in 1 spell from 2024-04-06T22:25:25Z, peak 3593 lbf (FS 2.78); did the alarm sound?
Alarm conditions in the logged data	Table A.8.2	info	not in this format's records; the logging system must show them
Tension display refresh rate	Table A.8.2	warn	no [monitoring] display_rate; must be 3 Hz or more
Tension measuring system calibration	Table A.8.2	warn	no [monitoring] last_calibration; every 6 months
Tension measuring accuracy	Table A.8.2	warn	no [monitoring] accuracy; must be within 4 % of the applied load
Sheaves and load-carrying rollers	Table A.8.2	warn	no [[sheave]] in the config (A.5.2.1 logs them)
Break test interval	Table A.8.2, A.5.1.1	pass	2023-11-15, due 2025-11-15; every 24 months
Assigned breaking load	A.5.6	pass	ABL 10000 lbf, 100 % of the NBL

The winch and tension member

Winch	hydro
Manufacturer	Markey
Model	DESH-5
Tension member (NSF identifier)	ASSUMED-HYDRO
Kind	cable
Description	assumed: 0.322 in. 3-conductor EM cable
Nominal breaking load (NBL)	10000 lbf
Assigned breaking load (ABL)	10000 lbf
Factor of safety, selected	2.5
Safe working tension (SWT = ABL / FS)	4000 lbf
Table	RVSS Appendix A Table A.8.2
Diameter (d)	0.322 in
Break test	2023-11-15 10200 lbf

Deployments

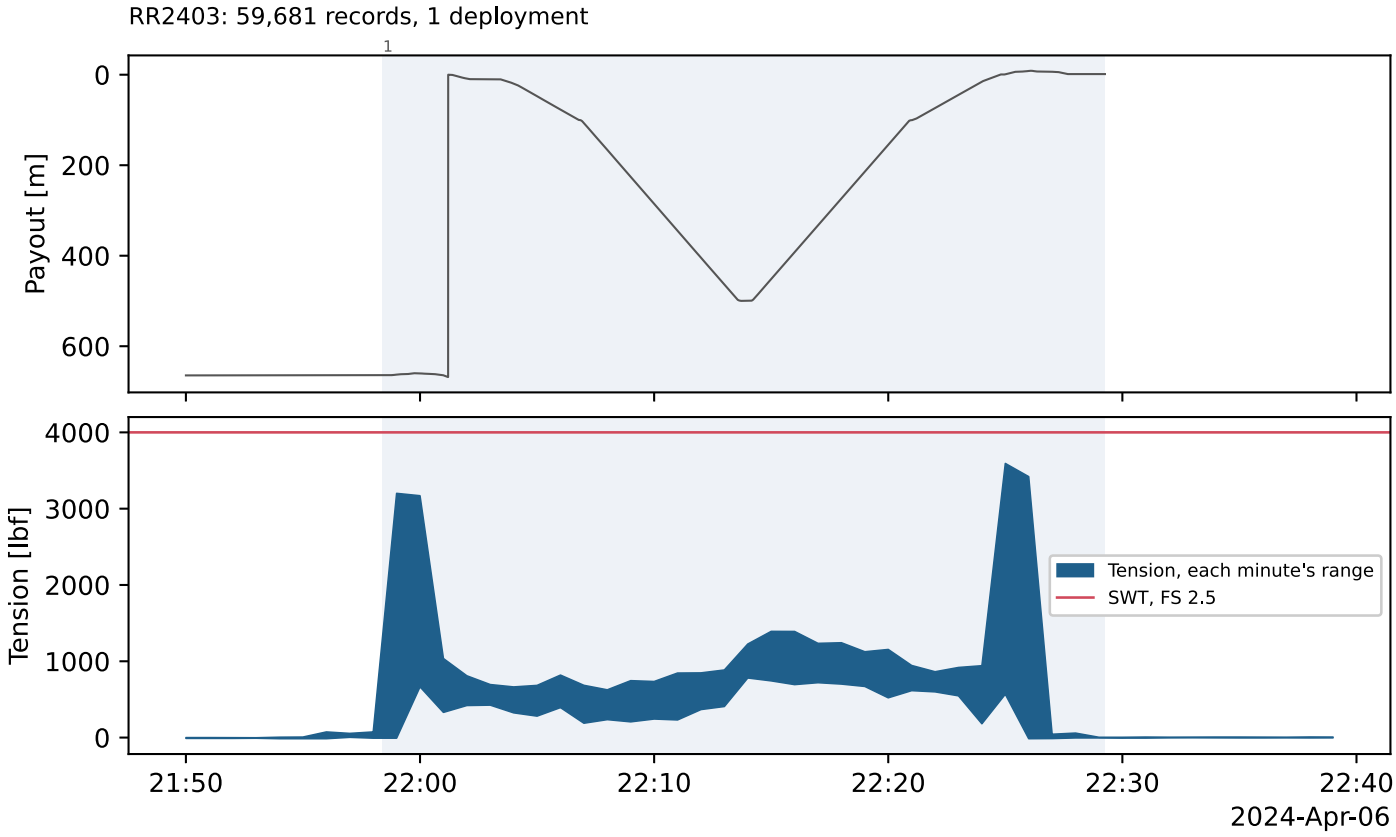
The wire log's deployments; the full log, with the time of each peak, is the _wire_log.csv beside this report. Each deployment includes its padding, the lift and the recovery.

#	Start (UTC)	End (UTC)	Duration [s]	Max payout [m]	Max tension [lbf]	Payout at max tension [m]	Min FS	Logging rate [Hz]	Longest gap [s]
1	2024-04-06 21:58:25	2024-04-06 22:29:15	1850.5	499.8	3593	-6.4	2.78	19.9	0.10

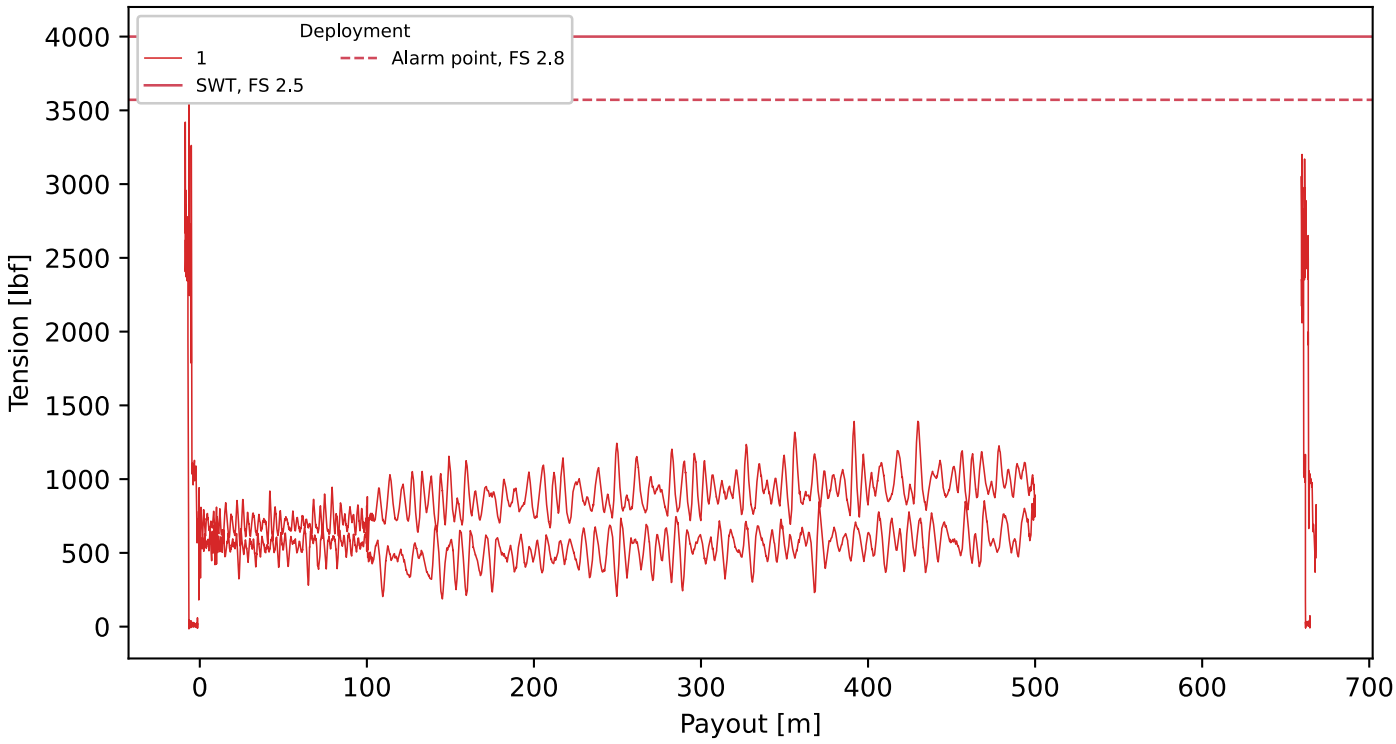
Records

Files read	1
Records read, within the cruise's dates	59,681
Lines skipped: bad checksum	23

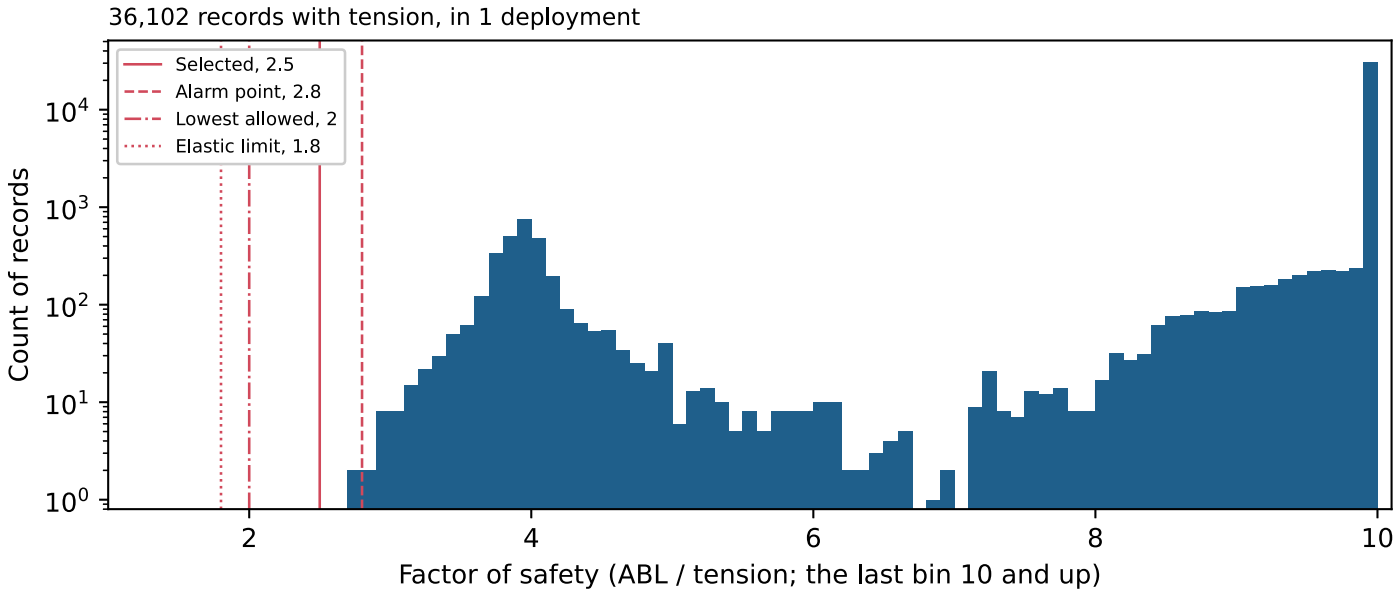
Timeline



Tension against payout



Factor of safety



Each deployment

Deployment 1

